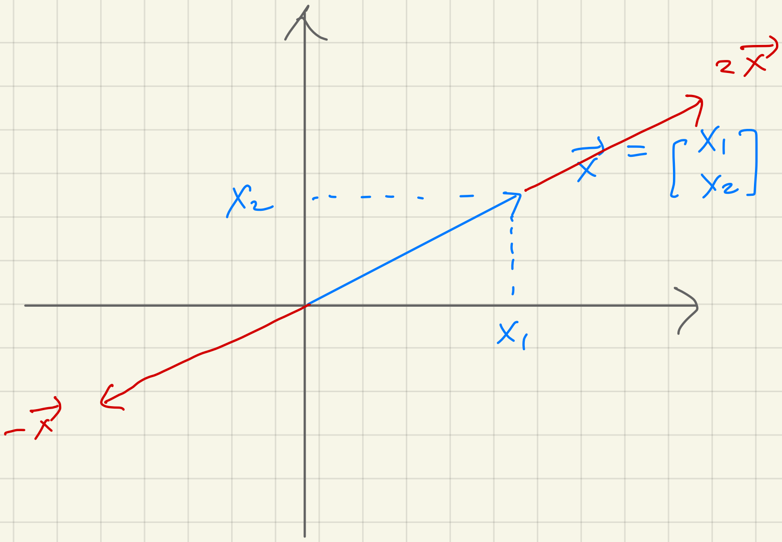
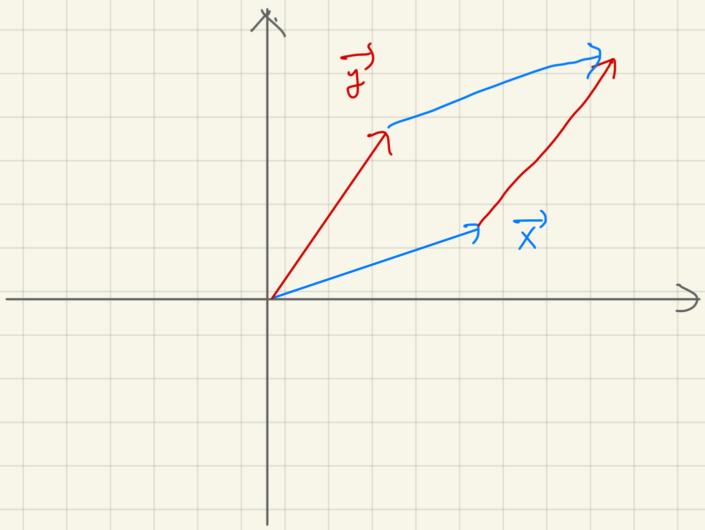


$2\vec{x}$

$-\vec{x}$

$\alpha \vec{x}$ $\alpha \in \mathbb{R}$

scalar multiplication



$\vec{x} + \vec{y}$ vector addition

$\shortparallel$

$\vec{y} + \vec{x}$

commutative

Set of column vectors:

$$\mathbb{R}^3 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\}$$

$\cdot : \mathbb{R} \times \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

- Scalar multiplication

$$\alpha \cdot \boldsymbol{x} = \alpha \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \alpha \cdot x_1 \\ \alpha \cdot x_2 \\ \alpha \cdot x_3 \end{bmatrix}, \quad \forall \alpha \in \mathbb{R},\ \forall \boldsymbol{x} \in \mathbb{R}^3$$

- Vector addition: $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$

$$\boldsymbol{x} + \boldsymbol{y} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix}, \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$$

- Commutativity of vector addition

$$\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{y} + \boldsymbol{x}, \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$$

- Associativity of vector addition

$$\boldsymbol{x} + (\boldsymbol{y} + \boldsymbol{z}) = (\boldsymbol{x} + \boldsymbol{y}) + \boldsymbol{z}, \quad \forall \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^3$$

- Identity element of vector addition

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \boldsymbol{x} + \mathbf{0} = \boldsymbol{x}, \quad \forall \boldsymbol{x} \in \mathbb{R}^3$$

- Inverse elements of vector addition

For each $\boldsymbol{x} \in \mathbb{R}^3$, $-\boldsymbol{x} = -1 \cdot \boldsymbol{x}$ satisfies $\boldsymbol{x} + (-\boldsymbol{x}) = \mathbf{0}$

- Compatibility of scalar multiplication with field multiplication

$$\alpha \cdot (\beta \cdot \boldsymbol{x}) = (\alpha \cdot \beta) \cdot \boldsymbol{x}, \quad \forall \alpha, \beta \in \mathbb{R},\ \forall \boldsymbol{x} \in \mathbb{R}^3$$

- Identity element of scalar multiplication

$$1 \cdot \boldsymbol{x} = \boldsymbol{x}, \quad \forall \boldsymbol{x} \in \mathbb{R}^3$$

- Distributivity of scalar multiplication w.r.t. vector addition

$$\alpha \cdot (\boldsymbol{x} + \boldsymbol{y}) = \alpha \cdot \boldsymbol{x} + \alpha \cdot \boldsymbol{y}, \quad \forall \alpha \in \mathbb{R},\ \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$$

- Distributivity of scalar multiplication w.r.t. field addition

$$(\alpha + \beta) \cdot \boldsymbol{x} = \alpha \cdot \boldsymbol{x} + \beta \cdot \boldsymbol{x}, \quad \forall \alpha, \beta \in \mathbb{R},\ \forall \boldsymbol{x} \in \mathbb{R}^3$$

Vector space:

A vector space over a field $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$) is a set V equipped with two binary operations (vector addition $+ : V \times V \to V$ and scalar multiplication $\cdot : \mathbb{F} \times V \to V$), which satisfies the following properties for all $\alpha, \beta \in \mathbb{F}$ and for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in V$.

- $\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{y} + \boldsymbol{x}$
- $\boldsymbol{x} + (\boldsymbol{y} + \boldsymbol{z}) = (\boldsymbol{x} + \boldsymbol{y}) + \boldsymbol{z}$
- $\exists\ \mathbf{0} \in V$ such that $\boldsymbol{x} + \mathbf{0} = \boldsymbol{x}$
- For each $\boldsymbol{x}$, there exists $-\boldsymbol{x}$ such that $\boldsymbol{x} + (-\boldsymbol{x}) = \mathbf{0}$
- $\alpha \cdot (\beta \cdot \boldsymbol{x}) = (\alpha \cdot \beta) \cdot \boldsymbol{x}$
- $1 \cdot \boldsymbol{x} = \boldsymbol{x}$
- $\alpha \cdot (\boldsymbol{x} + \boldsymbol{y}) = \alpha \cdot \boldsymbol{x} + \alpha \cdot \boldsymbol{y}$
- $(\alpha + \beta) \cdot \boldsymbol{x} = \alpha \cdot \boldsymbol{x} + \beta \cdot \boldsymbol{x}$

$\{x[n]\}_{n\in\mathbb{Z}} \cdot \{y[n]\}_{n\in\mathbb{Z}} = \sum_{n\in\mathbb{Z}} x[n]y[n]$

Discrete-time signals = sequences:

- A complex-valued sequence is a function from $\mathbb{Z}$ to $\mathbb{C}$.

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ 2D

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ 3D

$$\{x[n]\}_{n\in\mathbb{Z}} = \begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix}$$

- Scalar multiplication

$$\alpha \cdot \{x[n]\}_{n\in\mathbb{Z}} = \alpha \cdot \begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ \alpha \cdot x[-1] \\ \alpha \cdot x[0] \\ \alpha \cdot x[1] \\ \vdots \end{bmatrix}$$

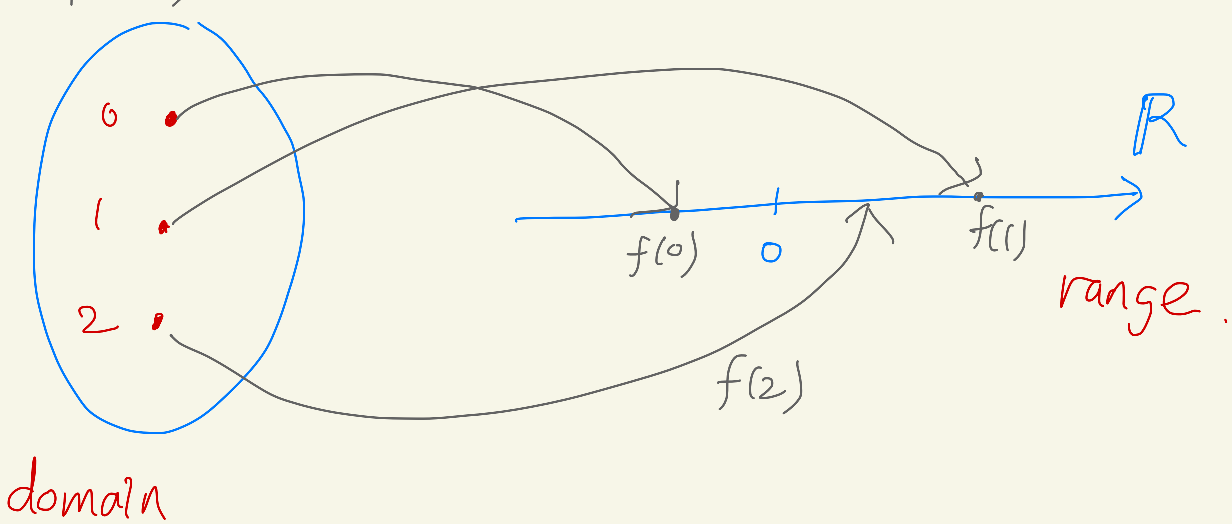
- Vector addition

$$\{x[n]\}_{n\in\mathbb{Z}}+\{y[n]\}_{n\in\mathbb{Z}} = \begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix} + \begin{bmatrix} \vdots \\ y[-1] \\ y[0] \\ y[1] \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ x[-1]+y[-1] \\ x[0]+y[0] \\ x[1]+y[1] \\ \vdots \end{bmatrix}$$

- Set of all complex-valued sequences is a vector space over scalar field $\mathbb{C}$

function is a mapping from a set (domain) to another set (range) that every element in the domain is uniquely assigned to an element in the range.

Example) $f : \{0, 1, 2\} \longrightarrow \mathbb{R}$



$\{x[n]\}_{n\in\mathbb{Z}} \cdot \{y[n]\}_{n\in\mathbb{Z}} = \sum_{n\in\mathbb{Z}} x[n]^* y[n]$

Alternative expressions:

- $\{x[n]\}_{n\in\mathbb{Z}}$ is a discrete-time complex-valued signal.

  $\iff$ x is the label for a mapping from $\mathbb{Z}$ to $\mathbb{C}$

  $\iff$ $\mathbb{Z} \ni n \mapsto x[n] \in \mathbb{C}$

- Scalar multiplication

$$\underbrace{\begin{bmatrix} \vdots \\ y[-1] \\ y[0] \\ y[1] \\ \vdots \end{bmatrix}}_{\{y[n]\}_{n\in\mathbb{Z}}} = \alpha \cdot \underbrace{\begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix}}_{\{x[n]\}_{n\in\mathbb{Z}}} = \begin{bmatrix} \vdots \\ \alpha \cdot x[-1] \\ \alpha \cdot x[0] \\ \alpha \cdot x[1] \\ \vdots \end{bmatrix}$$

$$\iff \quad y[n] = \alpha \cdot x[n], \quad \forall n \in \mathbb{Z}$$

- Vector addition

$$\underbrace{\begin{bmatrix} \vdots \\ z[-1] \\ z[0] \\ z[1] \\ \vdots \end{bmatrix}}_{\{z[n]\}_{n\in\mathbb{Z}}} = \underbrace{\begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix}}_{\{x[n]\}_{n\in\mathbb{Z}}} + \underbrace{\begin{bmatrix} \vdots \\ y[-1] \\ y[0] \\ y[1] \\ \vdots \end{bmatrix}}_{\{y[n]\}_{n\in\mathbb{Z}}} = \begin{bmatrix} \vdots \\ x[-1] + y[-1] \\ x[0] + y[0] \\ x[1] + y[1] \\ \vdots \end{bmatrix}$$

$$\iff \quad z[n] = x[n] + y[n], \quad \forall n \in \mathbb{Z}$$

Special discrete-time signals:

- There are signals represented by a few equations.
- Impulse: $\{\delta[n]\}_{n\in\mathbb{Z}}$

$$\delta[n] = \begin{cases} 1 & n = 0, \\ 0 & n \neq 0. \end{cases}$$

- Unit-step function: $\{u[n]\}_{n\in\mathbb{Z}}$

$$u[n] = \begin{cases} 1 & n \geq 0, \\ 0 & n < 0. \end{cases}$$

- Exponential signal: $\{a^n\}_{n\in\mathbb{Z}}$
- One-sided exponential signal

$$\begin{aligned} & x[n] = a^n \cdot u[n], \quad \forall n \in \mathbb{Z} \\ & \iff \{x[n]\}_{n\in\mathbb{Z}} = \{a^n\}_{n\in\mathbb{Z}} \odot \{u[n]\}_{n\in\mathbb{Z}} \\ & \iff x[n] = \begin{cases} a^n & n \geq 0, \\ 0 & n < 0. \end{cases} \end{aligned}$$

Hadamard product

= pointwise multiplication

a binary operator from $V \otimes V$ to V

$$\odot : V \times V \longrightarrow V$$
$$(\vec{x}, \vec{y}) \longmapsto \vec{x} \odot \vec{y}$$

Ex) $V = \mathbb{R}^3$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \odot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_1 \cdot b_1 \\ a_2 \cdot b_2 \\ a_3 \cdot b_3 \end{bmatrix} = \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Ex) $V = \{\text{sequences}\}$

$$\{z[n]\}_{n \in \mathbb{Z}} = \{x[n]\}_{n \in \mathbb{Z}} \odot \{y[n]\}_{n \in \mathbb{Z}}$$

$$z[n] = x[n] \cdot y[n] \quad \forall n \in \mathbb{Z}$$

Recall
$$x : \mathbb{Z} \longrightarrow \mathbb{C}$$
$$y : \mathbb{Z} \longrightarrow \mathbb{C}$$
$$z : \mathbb{Z} \longrightarrow \mathbb{C}$$

$\langle \cdot, \cdot \rangle$ inner product

$\{x(t)\}_{t\in\mathbb{R}} \cdot \{y(t)\}_{t\in\mathbb{R}} = \int x(t)^* y(t)\, dt$

$\langle \{x(t)\}_{t\in\mathbb{R}}, \{y(t)\}_{t\in\mathbb{R}} \rangle$

Continuous-time signals = functions on $\mathbb{R}$:

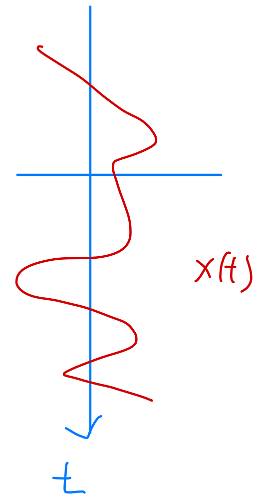
- $\{x(t)\}_{t\in\mathbb{R}}$ is a continuous-time complex-valued signal.
  $\iff$ x is the label for a mapping from $\mathbb{R}$ to $\mathbb{C}$
  $\iff$ $\mathbb{R} \ni t \mapsto x(t) \in \mathbb{C}$

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$\{x_k\}_{k=1,2,3}$

- Scalar multiplication

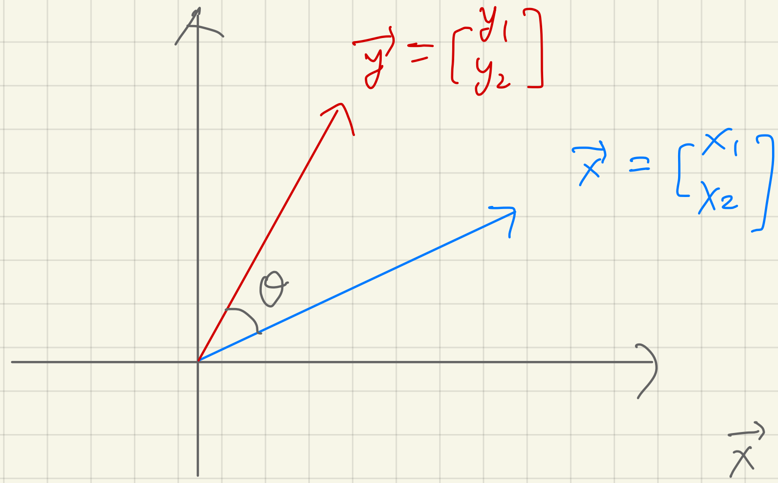
$$\{y(t)\}_{t\in\mathbb{R}} = \alpha \cdot \{x(t)\}_{t\in\mathbb{R}}$$
$$\iff y(t) = \alpha \cdot x(t), \quad \forall t \in \mathbb{R}$$



- Vector addition

$$\{z(t)\}_{t\in\mathbb{R}} = \{x(t)\}_{t\in\mathbb{R}} + \{y(t)\}_{t\in\mathbb{R}}$$
$$\iff z(t) = x(t) + y(t), \quad \forall t \in \mathbb{R}$$

- Set of all complex-valued functions on $\mathbb{R}$ is a vector space over scalar field $\mathbb{C}$



length of $\vec{x}$

$$\|\vec{x}\| = \sqrt{x_1^2 + x_2^2}$$

$$\vec{x} \cdot \vec{y} = x_1 y_1 + x_2 y_2$$

$$\vec{x} \cdot \vec{x} = \|\vec{x}\|^2$$

$$\cos\theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \cdot \|\vec{y}\|}$$

$$\cdot : V \times V \longrightarrow \mathbb{R}$$

$$(\vec{x}, \vec{y}) \longmapsto x_1 y_1 + x_2 y_2$$

$$a\vec{x} \cdot \vec{y} = a\,(\vec{x} \cdot \vec{y})$$

$$\vec{x} \cdot a\vec{y} = a\,(\vec{x} \cdot \vec{y})$$

$$(\vec{x} + \vec{z}) \cdot \vec{y} = \vec{x} \cdot \vec{y} + \vec{z} \cdot \vec{y}$$

$$\vec{x} \cdot (\vec{y} + \vec{z}) = \vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{z}$$

2D complex case

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad \vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \qquad x_1, x_2, y_1, y_2 \in \mathbb{C}$$

$$\vec{x} \cdot \vec{y} = x_1 y_1 + x_2 y_2 \ ?$$

$$z = Re(z) + j\, Im(z) \longleftrightarrow \begin{bmatrix} Re(z) \\ Im(z) \end{bmatrix}$$

$$\vec{x} \quad \begin{bmatrix} Re(x_1) \\ Im(x_1) \\ Re(x_2) \\ Im(x_2) \end{bmatrix} \qquad \vec{y} \quad \begin{bmatrix} Re(y_1) \\ Im(y_1) \\ Re(y_2) \\ Im(y_2) \end{bmatrix}$$

$$\vec{x} \cdot \vec{y} = Re(x_1)\, Re(y_1) + Im(x_1)\, Im(y_1) + Re(x_2)\, Re(y_2) + Im(x_2)\, Im(y_2)$$

$$Re[x_1^* y_1] = Re\{[Re(x_1) - j Im(x_1)]\,[Re(y_1) + j Im(y_1)]\}$$
$$= Re(x_1)\, Re(y_1) + Im(x_1)\, Im(y_1)$$
$$\cancel{-j\,(\qquad\qquad)}$$

$$\vec{x} \cdot \vec{y} = Re(x_1^* y_1) + Re(x_2^* y_2)$$
$$= Re(x_1^* y_1 + x_2^* y_2)$$

$$\vec{x} \cdot \vec{y} = \sum_{k=1,2} x_k^* y_k$$

$$\vec{x} \cdot \vec{x} = \sum_k x_k^* x_k = \sum_k |x_k|^2$$

Signals and systems:

$$H\{\{x(t)\}_{t\in\mathbb{R}}\} = \{y(t)\}_{t\in\mathbb{R}}$$
$$H\{x(t)\} = y(t)$$

- Continuous-time: $t \in \mathbb{R}$

$$\{x(t)\}_{t\in\mathbb{R}} \longrightarrow \boxed{\mathcal{H}_\mathsf{c}} \longrightarrow \{y(t)\}_{t\in\mathbb{R}}$$

signals: $\{x(t)\}_{t\in\mathbb{R}}$, $\{y(t)\}_{t\in\mathbb{R}}$
system: $\mathcal{H}_\mathsf{c}$

- Discrete-time: $n \in \mathbb{Z}$

$$\{x[n]\}_{n\in\mathbb{Z}} \longrightarrow \boxed{\mathcal{H}} \longrightarrow \{y[n]\}_{n\in\mathbb{Z}}$$

signals: $\{x[n]\}_{n\in\mathbb{Z}}$, $\{y[n]\}_{n\in\mathbb{Z}}$
system: $\mathcal{H}$

- Assumption:

  All signals are *complex valued*. Below, $j \triangleq \sqrt{-1}$:
  $x[n] = \mathrm{Re}(x[n]) + j\,\mathrm{Im}(x[n]) = |x[n]|\,e^{j\angle x[n]}$.

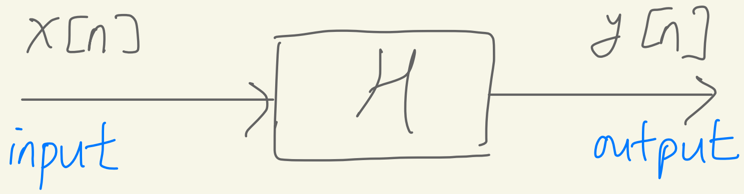
- Uniform sampling:

  We say that $\{x[n]\}_{n\in\mathbb{Z}}$ is a T-*sampled version* of $\{x(t)\}_{t\in\mathbb{R}}$ when $x[n] = x(nT)$ for every integer n. Here, T denotes the sampling interval in seconds.

- Key question:

  What "information" does the discrete-time sequence $\{x[n]\}_{n\in\mathbb{Z}}$ contain relative to the continuous-time waveform $\{x(t)\}_{t\in\mathbb{R}}$?

Convention in signal processing



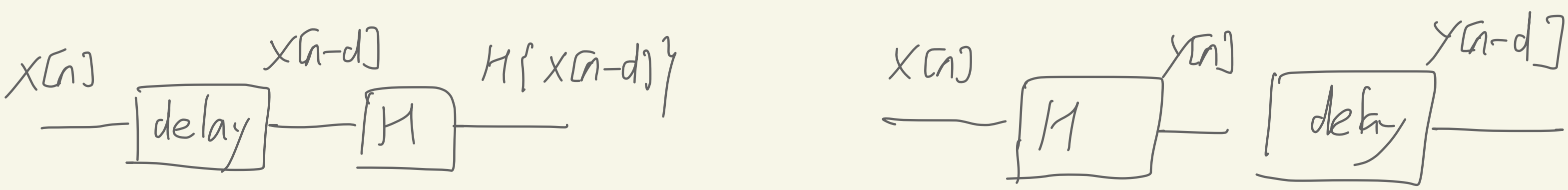
$$\{y[n]\} = H\{\{x[n]\}\}$$

Delay by d

$$H\{x[n]\} = x[n-d]$$

H is time invariant if

$$H\{x[n-d]\} = y[n-d]$$



for any d for any x[n]

A few important system properties:

For the statements below, assume that $\{y[n]\} = \mathcal{H}\{\{x[n]\}\}$.

1. Linear:
$$\mathcal{H}\{\underbrace{\alpha\{x[n]\} + \beta\{w[n]\}}_{\text{linear combination}}\} = \alpha\boxed{\mathcal{H}\{\{x[n]\}\}} + \beta\boxed{\mathcal{H}\{\{w[n]\}\}}$$
$$\forall\alpha, \beta \in \mathbb{C},\ \forall\{x[n]\}, \{w[n]\}$$

2. Time-invariant:
$$\mathcal{H}\{\{x[n-d]\}\} = \{y[n-d]\}, \quad \forall d \in \mathbb{Z},\ \forall\{x[n]\}$$

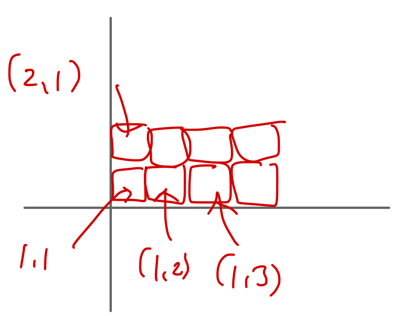
3. Causal:
At any time m, the output value $y[m]$ does not depend on the future input values $\{x[n]\}_{n>m}$.

4. BIBO Stable:
If $\{x[n]\}$ is bounded, then $\{y[n]\}$ will be bounded.

Recall that "bounded $\{x[n]\}$" means that there exists some finite value M_x such that $|x[n]| < M_x$ for all n.

2D signal

$x[m,n]$



index

(1,1) (1,2) (2,1)

total ordering

$\forall m,n \in \mathbb{Z}$ we can compare m and n so that $m=n$, $m>n$, $m<n$

Note: These properties can be directly re-stated for continuous-time systems as well.